

MULTIMODAL EMOTION RECOGNITION USING SPEECH AND TEXT

1. Introduction

Emotion recognition is an important task in human-computer interaction and artificial intelligence. This project aims to identify emotions using speech signals, text data, and multimodal fusion techniques.

The system predicts seven emotions:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Surprise
- Sad

Dataset used:

Toronto Emotional Speech Set (TESS)

2. Architecture Decisions

Speech Pipeline:

Audio

→ MFCC Feature Extraction

→ BiLSTM

→ Dense Layer

→ Emotion Prediction

Reason:

MFCC captures speech characteristics such as pitch and frequency, while BiLSTM captures temporal patterns.

Text Pipeline:

Text

→ DistilBERT

→ Dense Layer

→ Emotion Prediction

Reason:

DistilBERT captures contextual information from text efficiently.

Fusion Pipeline:

Speech Features

+

Text Features

→ Concatenation

→ Dense Layer

→ Emotion Prediction

Reason:

Fusion combines information from multiple modalities.

3. Experimental Results

Speech-only Accuracy:

99.82%

Text-only Accuracy:

12.5%

Fusion Accuracy:

36.6%

4. Analysis

Speech-only model achieved highest accuracy because emotional information is strongly present in speech signals.

Text-only model showed weak performance because TESS contains repeated neutral words with limited emotional meaning.

Fusion model improved over text-only performance but was affected by weak text features.

5. Confusion Matrix Analysis

Most emotional classes were classified correctly.

Minor confusion was observed between:

Fear and Angry

Happy and Angry

Surprise and Disgust

6. t-SNE Analysis

Distinct emotion clusters were observed.

The learned speech features created good separability among emotional classes.

7. Future Improvements

- Real-time prediction
- Attention-based fusion
- Better textual datasets
- Streamlit deployment